



Integration.

In this lecture, we will introduce the concept of integration in mathematics, then we will talk about integration fundamentals and some basic integration formulas. Moreover, we will discuss the idea of using Integration Techniques for more complicated integrals, and we will elaborate on the first of the techniques that will be studied throughout this course, the Integration by Substitution also known as u-substitution or change of variables. Furthermore, we will be examining the Integration by Parts, which is a technique used to evaluate integrals of products of functions. Finally, we will be examining the Integration of Rational Functions by Partial Fractions.

5.1 Integration Definition.

Given a function $f(x)$, an anti-derivative $F(x)$ is any function such that,

$$F'(x) = f(x)$$

the most general anti-derivative of $f(x)$ is called an indefinite integral,

$$\int f(x)dx = F(x) + c$$

where $\int$ is the integral symbol, $f(x)$ is the integrand, x is the integration variable, and the c is the constant of integration.

Remark 5.1.

1. Often it is just said integral instead of indefinite integral (or definite integral). It will be clear from the context of the problem.
2. The process of finding the indefinite integral is called integration or integrating $f(x)$. If we need to be specific about the integration variable, we say the integrating of $f(x)$ with respect to x .
3. Some integrals can be calculated by merely recollecting which function has which derivative. However, the integration techniques mainly involve classifying the functions according to which manipulation types that change the function into a form with more easily recognized antiderivatives.



5.2 Integration Formulas

In the following integrations formulas, u will represent the function $u(x)$, and du is the derivative with respect to x ,

Integration Formula.

1. $\int du = u + c$

2. $\int kdu = ku + c$

k is any constant

3. $\int u^n du = \frac{1}{n+1} u^{n+1} + c$

$n \neq -1$

4. $\int \frac{1}{u} du = \ln(u) + c$

Exponential Functions.

5. $\int e^x dx = e^x + c$

6. $\int e^{kx} dx = \frac{1}{k} e^x + c$

7. $\int e^u du = e^u + c$

8. $\int a^x dx = \frac{1}{\ln(a)} a^x + c$

$a > 0, a \neq 1$

9. $\int a^{kx} dx = \frac{1}{k \ln(a)} a^{kx} + c$

$a > 0, a \neq 1$

10. $\int a^u du = \frac{1}{\ln(a)} a^u + c$

$a > 0, a \neq 1$

Trigonometric Functions.

11. $\int \sin(u) du = -\cos(u) + c$

12. $\int \cos(u) du = \sin(u) + c$

13. $\int \sec^2(u) du = \tan(u) + c$

14. $\int \csc^2(u) du = -\cot(u) + c$

15. $\int \sec(u) \tan(u) du = \sec(u) + c$

16. $\int \csc(u) \cot(u) du = -\csc(u) + c$



Inverse Trigonometric Functions.

17. $\int \frac{1}{\sqrt{a^2 - u^2}} du = \sin^{-1} \left(\frac{u}{a} \right) + c$
18. $\int \frac{-1}{\sqrt{a^2 - u^2}} du = \cos^{-1} \left(\frac{u}{a} \right) + c$
19. $\int \frac{1}{a^2 + u^2} du = \frac{1}{a} \tan^{-1} \left(\frac{u}{a} \right) + c = -\frac{1}{a} \cot^{-1} \left(\frac{u}{a} \right) + c$
20. $\int \frac{1}{u\sqrt{u^2 - a^2}} du = \frac{1}{a} \sec^{-1} \left(\frac{u}{a} \right) + c = -\frac{1}{a} \csc^{-1} \left(\frac{u}{a} \right) + c$

Hyperbolic Functions.

21. $\int \sinh(u) du = \cosh(u) + c$
22. $\int \cosh(u) du = \sinh(u) + c$
23. $\int \operatorname{sech}^2(u) du = \tanh(u) + c$
24. $\int \operatorname{csch}^2(u) du = -\coth(u) + c$
25. $\int \operatorname{sech}(u) \tanh(u) du = -\operatorname{sech}(u) + c$
26. $\int \operatorname{csch}(u) \coth(u) du = -\operatorname{csch}(u) + c$

Inverse Hyperbolic Functions.

27. $\int \frac{1}{\sqrt{u^2 + a^2}} du = \sinh^{-1} \left(\frac{u}{a} \right) + c$
28. $\int \frac{1}{\sqrt{u^2 - a^2}} du = \cosh^{-1} \left(\frac{u}{a} \right) + c$
29. $\int \frac{1}{a^2 - u^2} du = \frac{1}{a} \tanh^{-1} \left(\frac{u}{a} \right) + c$ $-a < u < a$
30. $\int \frac{1}{a^2 + u^2} du = \frac{1}{a} \coth^{-1} \left(\frac{u}{a} \right) + c$ $u < -a \text{ or } u > a$
31. $\int \frac{1}{u\sqrt{a^2 - u^2}} du = -\frac{1}{a} \operatorname{sech}^{-1} \left(\frac{u}{a} \right) + c$
32. $\int \frac{1}{u\sqrt{a^2 + u^2}} du = -\frac{1}{a} \operatorname{csch}^{-1} \left(\frac{u}{a} \right) + c$

Example 5.1 Solve the following integrals.

$$1. \int dx = x + c$$

$$2. \int 7dx = 7 \int dx \\ = 7x + c$$

$$3. \int x^7 dx = \frac{1}{7+1} x^{7+1} + c \\ = \frac{1}{8} x^8 + c$$

$$4. \int (x^7 - 5x^3 + 1) dx = \int x^7 dx - \int 5x^3 dx + \int dx \\ = \frac{1}{8} x^8 - 5 \int x^3 dx + x + c \\ = \frac{1}{8} x^8 - 5 \left(\frac{1}{4} x^4 \right) + x + c$$

$$5. \int \frac{7}{x} dx = 7 \int \frac{1}{x} dx \\ = 7 \ln(x) + c$$

$$6. \int 7^{5x} dx = \frac{1}{5} \frac{1}{\ln(7)} 7^{5x} + c$$

$$7. \int 3^{\sin(x)} \cos(x) dx = \frac{1}{\ln(3)} 3^{\sin(x)} + c$$

$$8. \int \frac{1}{\sqrt{9-x^2}} dx = \int \frac{1}{\sqrt{3^2-x^2}} dx \\ = \sin^{-1} \left(\frac{x}{3} \right) + c$$

$$9. \int \frac{-5}{\sqrt{49-x^2}} dx = 5 \int \frac{-1}{\sqrt{7^2-x^2}} dx \\ = 5 \cos^{-1} \left(\frac{x}{7} \right) + c$$

$$10. \int \frac{1}{\sqrt{x^2+7}} du = \sinh^{-1} \left(\frac{x}{\sqrt{7}} \right) + c$$



Example 5.2 Show that $\int \frac{1}{\sqrt{a^2 - u^2}} du = \sin^{-1} \left(\frac{u}{a} \right) + c$.

Let $y = \sin^{-1} \left(\frac{u}{a} \right)$,

$$\sin(y) = \frac{u}{a} \rightarrow a \sin(y) = u.$$

Using implicit differentiation, we obtain

$$a \cos(y) \frac{dy}{du} = 1$$

$$\frac{dy}{du} = \frac{1}{a \cos(y)}$$

Now, for $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$, $\cos(y) \geq 0$. We apply the Pythagorean identity,

$$\sin^2(y) + \cos^2(y) = 1$$

$$\cos(y) = \sqrt{1 - \sin^2(y)}$$

This gives,

$$\frac{dy}{du} = \frac{1}{a \cos(y)}$$

$$= \frac{1}{a \sqrt{1 - \sin^2(y)}}$$

$$= \frac{1}{\sqrt{a^2 - a^2 \sin^2(y)}}$$

$$= \frac{1}{\sqrt{a^2 - u^2}}$$

Then for $-a \leq u \leq a$. Integrate both sides w.r.t u ,

$$\int \frac{1}{\sqrt{a^2 - u^2}} du = \int \frac{dy}{du} du$$

$$= y + c$$

$$= \sin^{-1} \left(\frac{u}{a} \right) + c$$



5.3 Integration Techniques

Many integration formulas can be derived directly from their corresponding derivative formulas since it will be apparent that the function to be integrated is a derivative in some straightforward way since more complicated integrals require extra work.

By studying the techniques, such as Integration by substitution, Integration by parts, Integrals Involving Trigonometric Functions, Integration by Trigonometric Substitution, Integration of Rational Functions by Partial Fractions, Integrals Involving Roots, Integrals Involving Quadratics. You will be able to solve a greater variety of applied calculus problems.

5.3.1 Integration by Substitution

In calculus, integration by substitution, also known as u-substitution or change of variables, is a method for evaluating integrals. Direct application of the fundamental theorem of calculus to find an antiderivative can be quite tricky, and integration by substitution can help simplify that task.

The point of substitution is to make the integration step easy. The "work" involved in making the proper substitution depends mainly on experience as a guide.

$$\begin{aligned} I &= \int \hat{f}(g(x))\dot{g}(x)dx \\ &= \int \hat{f}(u)du \\ &= f(u) + c = f(g(x)) + c \end{aligned}$$

The method works by transforming the integral into one of the standard forms but in other variable say u , and then evaluate it with respect to that variable and then change the result back.

Following is simple steps one can carry to apply this method,

1. Substitute $u = g(x)$ and hence $du = g'(x)dx$.
2. Integrate with respect to u .
3. Replace u by $g(x)$ in the result.

To gain experience, we now embark on many examples.

Examples 5.3 Evaluate the following integrals.

1. $I = \int (x^3 + 2x^2 - 5x + 5)^3 (9x^2 + 12x - 15) dx$

Let $u = x^3 + 2x^2 - 5x + 5$

and hence $du = (3x^2 + 4x - 5)dx$. The integral becomes,

$$\begin{aligned} I &= \int (x^3 + 2x^2 - 5x + 5)^3 3(3x^2 + 4x - 5) dx \\ &= 3 \int u^3 du = 3 \frac{u^{3+1}}{3+1} + c = \frac{3}{4} u^4 + c \\ &= \frac{3}{4} (x^3 + 2x^2 - 5x + 5)^4 + c \end{aligned}$$

2. $I = \int x\sqrt{1+x^2} dx$

Let $u = 1 + x^2$

and hence $du = 2x dx$. The integral becomes,

$$\begin{aligned} I &= \frac{1}{2} \int \sqrt{1+x^2} 2x dx \\ &= \frac{1}{2} \int \sqrt{u} du = \frac{1}{2} \int u^{\frac{1}{2}} du \\ &= \frac{1}{2} \frac{u^{\frac{1}{2}+1}}{\frac{1}{2}+1} + c = \frac{1}{2} \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + c = \frac{1}{3} u^{\frac{3}{2}} + c \\ &= \frac{1}{3} (1+x^2)^{\frac{3}{2}} + c \end{aligned}$$



3. $I = \int \cos(7x + 5) dx$

Let $u = 7x + 5$

and hence $du = 7 dx$. The integral becomes,

$$\begin{aligned} I &= \frac{1}{7} \int \cos(7x + 5) 7 dx \\ &= \frac{1}{7} \int \cos(u) du = \frac{1}{7} \sin(u) + c \\ &= \frac{1}{7} \sin(7x + 5) + c \end{aligned}$$

4. $I = \int x^2 \sin(x^3) dx$

Let $u = x^3$

and hence $du = 3x^2 dx$. The integral becomes,

$$\begin{aligned} I &= \frac{1}{3} \int \sin(x^3) 3x^2 dx \\ &= \frac{1}{3} \int \sin(u) du = \frac{1}{3} (-\cos(u)) + c \\ &= -\frac{1}{3} \cos(u) + c \\ &= -\frac{1}{3} \cos(x^3) + c \end{aligned}$$

5. $I = \int \frac{\sec^2(x)}{\tan(x)} dx$

Let $u = \tan(x)$

and hence $du = \sec^2(x) dx$. The integral becomes,

$$\begin{aligned} I &= \int \frac{1}{\tan(x)} \sec^2(x) dx \\ &= \int \frac{1}{u} du = \ln(u) + c \\ &= \ln(\tan(x)) + c \end{aligned}$$

$$6. I = \int \frac{1}{\sqrt{x}(1+\sqrt{x})^2} dx$$

$$\text{Let } u = 1 + \sqrt{x}$$

and hence $du = \frac{1}{2\sqrt{x}} dx$. The integral becomes,

$$I = 2 \int \frac{1}{(1+\sqrt{x})^2} \frac{1}{2\sqrt{x}} dx$$

$$= 2 \int \frac{1}{u^2} du$$

$$= 2 \int u^{-2} du$$

$$= 2 \frac{u^{-2+1}}{-2+1} + c$$

$$= -2 \frac{1}{u} + c$$

$$= \frac{-2}{1+\sqrt{x}} + c$$

$$7. I = \int \left(1 - \frac{1}{x}\right) \cos(x - \ln x) dx$$

$$\text{Let } u = x - \ln x$$

and hence $du = \left(1 - \frac{1}{x}\right) dx$. The integral becomes,

$$I = \int \cos(u) du$$

$$= \sin(u) + c$$

$$= \sin(x - \ln x) + c$$

$$8. I = \int \frac{18 \tan^2(x) \sec^2(x)}{[2 + \tan^3(x)]^2} dx.$$

$$\text{Let } u = 2 + \tan^3(x)$$

and hence $du = 3 \tan^2(x) \sec^2(x) dx$. The integral becomes,

$$I = \int \frac{6}{[2 + \tan^3(x)]^2} 3 \tan^2(x) \sec^2(x) dx$$

$$= 6 \int \frac{1}{u^2} du$$

$$= 6 \frac{u^{-2+1}}{-2+1} + c$$

$$= \frac{-6}{u} + c$$

$$= \frac{-6}{2 + \tan^3(x)} + c$$

$$9. I = \int \sin(1-x)(2 - \cos(1-x))^4 dx \quad \text{H.W.}$$

$$10. I = \int \frac{2x^3+1}{(x^4+2x)^3} dx \quad \text{H.W.}$$

5.3.2 Integration by Parts.

It is a process that finds the integral of function's product in terms of the integral of their derivative and antiderivative product. It is frequently used to transform the integral of a product of functions into an integral with a more convenient solution. To use this formula, we will need to recognize a choice for u and dv , then compute du by differentiation, and v by integration. After that, we apply the following integration by parts formula,

$$I = \int u dv = uv - \int v du$$

This above formula expresses one integral $\int u dv$, in terms of a second integral $\int v du$, with a proper choice of u and dv , the second integral may be easier to evaluate than the first.

Remark 5.2.

Various choices may be available for u and dv , but one may choose the following suggested priority for the choice of u as,

$$1^{\text{st}} \ln(x), 2^{\text{nd}} x^n, 3^{\text{rd}} e^{kx},$$

and the rest is dv .

5.3.3 Tabular integration by parts

It is a special technique for integration by parts that can be applied to a particular product of functions where one can be differentiated multiple times with ease say $f(x)$, and the other function can be integrated multiple times with ease say $g(x)$.

$$I = \int f(x)g(x) dx$$

To apply the method, we need to construct the table below, with an index starting from zero as the first column, then sign alternating starting with + then – then + until the last row is reached as the second column, after that $f(x)$ and its subsequent derivatives until zero is reached as third column "A," furthermore, "B" with $g(x)$ and its subsequent integrals until the size of B is the same as that of A.

Index	sign	A	B
0	+	$f(x)$	$g(x)$
1	-	$\dot{f}(x)$	$\int g(x) dx$
2	+	$\ddot{f}(x)$	$\int \int g(x) dx$
3	-	$\dddot{f}(x)$	$\int \int \int g(x) dx$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	$(-1)^n$	0	



For the complete result of the integral, we need the summation until $n - 1$ is reached, of the product of the entries in row i of sign, A , and row $i + 1$ of B .

$$I = f(x) \int g(x) dx - f'(x) \int \int g(x) dx + f''(x) \int \int \int g(x) dx - \dots + c$$

Example 5.4 Evaluate the following integrals.

1. $I = \int x^2 \ln(x) dx$

Let, $u = \ln(x) \rightarrow du = \frac{1}{x} dx$

and, $dv = x^2 dx \rightarrow v = \int x^2 dx = \frac{1}{3} x^3$

Now, according to the integration by parts formula, the integral becomes,

$$\int x^2 \ln(x) dx = \int u dv$$

$$= uv - \int v du$$

$$= \ln(x) \frac{1}{3} x^3 - \int \frac{1}{3} x^3 \frac{1}{x} dx$$

$$= \frac{1}{3} x^3 \ln(x) - \frac{1}{3} \int x^2 dx$$

$$= \frac{1}{3} x^3 \ln(x) - \frac{1}{3} \left(\frac{1}{3} x^3 \right) + c$$

$$= \frac{1}{3} x^3 \ln(x) - \frac{1}{9} x^3 + c$$

2. $I = \int x \sin(x) dx$

let, $u = x \rightarrow du = dx$

and, $dv = \sin(x) dx \rightarrow v = \int \sin(x) dx = -\cos(x)$

Now, the integral becomes,

$$\int x \sin(x) dx = \int u dv$$

$$= uv - \int v du$$

$$= -x \cos(x) + \int \cos(x) dx$$

$$= -x \cos(x) + \sin(x) + c$$



3. $I = \int e^x \cos(x) dx$

Let, $u = e^x \rightarrow du = e^x dx$

and then, $dv = \cos(x) dx \rightarrow v = \int \cos(x) dx = \sin(x)$

hence the integral becomes,

$$I = \int u dv$$

$$= e^x \sin(x) - \int e^x \sin(x) dx$$

$$= e^x \sin(x) - J$$

But now, we should evaluate J in order to evaluate I

$$J = \int e^x \sin(x) dx$$

Let, $u = e^x \rightarrow du = e^x dx$

and then, $dv = \sin(x) dx \rightarrow v = \int \sin(x) dx = -\cos(x)$

hence the integral becomes,

$$J = \int u dv$$

$$= -e^x \cos(x) + \int \cos(x) e^x dx$$

$$= -e^x \cos(x) + I$$

and the all over Integral becomes,

$$I = e^x \sin(x) - (-e^x \cos(x) + I)$$

$$= e^x \sin(x) + e^x \cos(x) - I$$

$$2I = e^x \sin(x) + e^x \cos(x)$$

$$I = \frac{1}{2} e^x [\sin(x) + \cos(x)]$$



$$4. I = \int \frac{1}{x^2} \ln^2(x) dx$$

$$\text{Let, } u = \ln^2(x) \rightarrow du = 2 \ln(x) \frac{1}{x} dx$$

$$\text{and, } dv = \frac{1}{x^2} dx \rightarrow v = -\frac{1}{x}$$

$$I = uv - \int v du$$

$$= -\frac{1}{x} \ln^2(x) - \int \left(-\frac{1}{x}\right) \left(2 \ln(x) \frac{1}{x}\right) dx$$

$$= -\frac{1}{x} \ln^2(x) - \int \left(-\frac{1}{x}\right) \left(2 \ln(x) \frac{1}{x}\right) dx$$

$$= -\frac{1}{x} \ln^2(x) + 2 \int \frac{1}{x^2} \ln(x) dx$$

$$= -\frac{1}{x} \ln^2(x) + 2J$$

Use integration by parts again.

$$\text{Let, } u = \ln(x) \rightarrow du = \frac{1}{x} dx$$

$$\text{and, } dv = \frac{1}{x^2} dx \rightarrow v = -\frac{1}{x}$$

$$J = -\frac{1}{x} \ln(x) - \int \left(-\frac{1}{x}\right) \left(\frac{1}{x}\right) dx$$

$$= -\frac{1}{x} \ln(x) + \int \frac{1}{x^2} dx$$

$$= -\frac{1}{x} \ln(x) - \frac{1}{x}$$

Therefore,

$$I = -\frac{1}{x} \ln^2(x) + 2 \left[-\frac{1}{x} \ln(x) - \frac{1}{x} \right] + C$$

$$= -\frac{1}{x} \ln^2(x) - \frac{2}{x} \ln(x) - \frac{2}{x} + C$$

$$= -\frac{1}{x} [\ln^2(x) + 2 \ln(x) + 2] + C$$



5. $I = \int x^2 e^{3x} dx$

Let, $u = x^2 \rightarrow du = 2x dx$

and then, $dv = e^{3x} dx \rightarrow v = \int e^{3x} dx = \frac{1}{3} e^{3x}$,

hence the integral becomes,

$$I = \int u dv$$

$$= uv - \int v du$$

$$= x^2 \frac{1}{3} e^{3x} - \int \frac{1}{3} e^{3x} 2x dx$$

$$= \frac{1}{3} x^2 e^{3x} - \frac{2}{3} \int x e^{3x} dx = \frac{1}{3} x^2 e^{3x} - \frac{2}{3} J$$

But now, we should evaluate J in order to evaluate I

$$J = \int x e^{3x} dx$$

Let, $u = x \rightarrow du = dx$

and then, $dv = e^{3x} dx \rightarrow v = \int e^{3x} dx = \frac{1}{3} e^{3x}$

hence the integral becomes,

$$J = \int u dv$$

$$= uv - \int v du$$

$$= \frac{1}{3} x e^{3x} - \frac{1}{3} \int e^{3x} dx$$

$$= \frac{1}{3} x e^{3x} - \frac{1}{9} e^{3x}$$

and the all over Integral becomes,

$$I = \int x^2 e^{3x} dx$$

$$= \frac{1}{3} x^2 e^{3x} - \frac{2}{3} \left(\frac{1}{3} x e^{3x} - \frac{1}{9} e^{3x} \right) + c$$

$$= \frac{1}{3} x^2 e^{3x} - \frac{2}{9} x e^{3x} + \frac{2}{27} e^{3x} + c$$

6. $I = \int x^4 e^{-x} dx$

Let, $u = f(x) = x^4$,

and, $dv = g(x) = e^{-x} dx$

Now, construct the table as follows,

Index	sign	A	B
0	+	x^4	e^{-x}
1	-	$4x^3$	$-e^{-x}$
2	+	$12x^2$	e^{-x}
3	-	$24x$	$-e^{-x}$
4	+	24	e^{-x}
5	-	0	$-e^{-x}$

The complete result of the integral, is given by,

$$I = -x^4 e^{-x} - 4x^3 e^{-x} - 12x^2 e^{-x} - 24x e^{-x} - 24e^{-x} + c$$

$$= -(x^4 + 4x^3 + 12x^2 + 24x + 24)e^{-x} + c$$

7. $\int x^3 \sin(x) dx$ H.W.3

8. $\int e^{5x} \sin(3x) dx$ H.W.4

5.3.4 Integrating Rational Function by Partial Fractions.

The method of partial fraction decomposition allows us to convert the rational functions into sums of more straightforward, and easily integrated rational functions. We will take a look at fractions where the denominator is a product of linear factors with no repeats. Furthermore, we will examine the case of linear factors with repeats.



Also, quadratic factors which cannot be factored into linear factors, but have no repeats, in addition to the case where the denominator has irreducible quadratic factors and repeats. **A rational function is a quotient of two polynomial functions** $p_n(x)$, and $q_m(x)$ where n , and m are the degrees of the polynomials, i.e., A rational should have the form,

$$f(x) = \frac{p_n(x)}{q_m(x)}$$

provided that $q_m(x) \neq 0$, and $n < m$. If $n \geq m$ then we perform long division before we can use partial fraction decomposition.

Example 5.5 Consider the following rational function,

$$f(x) = \frac{p_n(x)}{q_m(x)} = \frac{x^5 - 2x^4 + 3x^2 - x + 4}{x^2 - 3x + 4}$$

here we notice that the degree of the polynomial $p_n(x)$ in the numerator is $n = 5$ and the degree of the polynomial $q_m(x)$ in the denominator is $m = 2$, so when compared we have $n = 5 > m = 2$, and hence we perform long division a long division as follows,

$$\begin{array}{r} x^3 + x^2 - x - 4 \\ x^2 - 3x + 4 \overline{) x^5 - 2x^4 + 3x^2 - x + 4} \\ \underline{x^5 - 3x^4 + 4x^3} \\ x^4 - 4x^3 + 3x^2 - x + 4 \\ \underline{x^4 - 3x^3 + 4x^2} \\ -x^3 - x^2 - x + 4 \\ \underline{-x^3 + 3x^2 - 4x} \\ -4x^2 + 3x + 4 \\ \underline{-4x^2 + 12x - 16} \\ -9x + 20 \end{array}$$

and the result would be as follows,

$$f(x) = (x^3 + x^2 - x - 4) + \frac{-9x + 20}{x^2 - 3x + 4}$$

Now, going back to our method, we have the aim of evaluating the following integral, provided that $q_m(x) \neq 0$, and $n < m$,

$$\int f(x) dx = \int \frac{p_n(x)}{q_m(x)} dx$$

and this aim could be achieved by carrying out the following steps,

1. We factor out the polynomial $q_m(x)$ in the denominator, and accordingly, we have three cases, comes either as one at a time, or as a mix of two, or three,

a. If the roots are non-repeated, say $x - \alpha_1, x - \alpha_2, \dots, x - \alpha_\beta$, then we write the partial fractions as follows,

$$\frac{A_1}{x - \alpha_1} + \frac{A_2}{x - \alpha_2} + \dots + \frac{A_\beta}{x - \alpha_\beta}$$

where $A_1, A_2, \dots, A_\beta$ are constant to be evaluated.

b. If some of the roots are repeated, say $(x - \alpha_1)^\beta$ is the highest power of $x - \alpha_1$, then to this factor, we assign the sum of β partial fractions, as follows,

$$\frac{B_1}{x - \alpha_1} + \frac{B_2}{(x - \alpha_1)^2} + \dots + \frac{B_\beta}{(x - \alpha_1)^\beta}$$

where $B_1, B_2, \dots, B_\beta$ are constant to be evaluated.

c. If we have the root of a Quadratic term of the form $x^2 + ax + b$, which cannot be factored, we assign the following partial fraction to it,

$$\frac{Cx + D}{x^2 + ax + b}$$

where C and D are constant to be evaluated.

Now, we can summarize these cases in the following table,



Denominator	Integrand	Partial Fractions
Non-Repeated	$\frac{p_n(x)}{(x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_\beta)}$	$\frac{A_1}{x - \alpha_1} + \frac{A_2}{x - \alpha_2} + \cdots + \frac{A_\beta}{x - \alpha_\beta}$
Repeated	$\frac{p_n(x)}{(x - \alpha)^\beta}$	$\frac{B_1}{x - \alpha} + \frac{B_2}{(x - \alpha)^2} + \cdots + \frac{B_\beta}{(x - \alpha)^\beta}$
Quadratic terms	$\frac{p_n(x)}{x^2 + ax + b}$	$\frac{Cx + D}{x^2 + ax + b}$

2. We replace the rational function inside the integral with the new partial fractioned function.
3. Evaluate the new simpler integrals using the formulas or the other methods.

Example 5.6 Evaluate the following integrals.

1. $I = \int \frac{x^2 + 3x + 1}{x^3 - x} dx$

Factor out $x^3 - x$,

$$x^3 - x = x(x^2 - 1) = x(x - 1)(x + 1)$$

Put it in the denominator,

$$\begin{aligned} I &= \int \frac{x^2 + 3x + 1}{x^3 - x} dx \\ &= \int \frac{x^2 + 3x + 1}{x(x - 1)(x + 1)} dx \end{aligned}$$

Apply the partial fraction decomposition,

$$\begin{aligned} \frac{x^2 + 3x + 1}{x(x - 1)(x + 1)} &= \\ &= \frac{A_1}{x} + \frac{A_2}{x - 1} + \frac{A_3}{x + 1} \\ &= \frac{A_1(x - 1)(x + 1) + A_2x(x + 1) + A_3x(x - 1)}{x(x - 1)(x + 1)} \end{aligned}$$

Equate the numerators from both sides to have the main equation that needs to be solved to determine the coefficients.

$$x^2 + 3x + 1 = A_1(x - 1)(x + 1) + A_2x(x + 1) + A_3x(x - 1)$$

There are two different strategies for finding the coefficients. One, we will call equating coefficients method and the other, substitution method.

Method One (equating coefficients),

Expand the R.H.S. of the main equation, gather the constants for each different degree of x , compare the coefficients at both sides to construct a set of simultaneous equations, and finally solve them to find the constants,

$$\begin{aligned} x^2 + 3x + 1 &= A_1(x - 1)(x + 1) + A_2x(x + 1) + A_3x(x - 1) \\ &= A_1x^2 - A_1 + A_2x^2 + A_2x + A_3x^2 - A_3x \\ &= (A_1 + A_2 + A_3)x^2 + (A_2 - A_3)x - A_1 \end{aligned}$$

$$A_1 + A_2 + A_3 = 1 \tag{1}$$

$$A_2 - A_3 = 3 \tag{2}$$

$$-A_1 = 1 \rightarrow A_1 = -1 \tag{3}$$

Now, substitute (3) into (1), we obtain,

$$A_2 + A_3 = 2 \tag{4}$$

and by adding (2) to (4), we have,

$$2A_2 = 5 \rightarrow A_2 = \frac{5}{2} \tag{5}$$

substitute (5) into (4), we obtain,

$$\frac{5}{2} + A_3 = 2 \rightarrow A_3 = 2 - \frac{5}{2} = -\frac{1}{2}$$

Method Two (substitution),

If we set up the decomposition correctly, then there must be coefficients' values that satisfy Equation for all values of x . i.e., this equation is valid for any value of x we substitute into it. Hence, if we choose the values plugged into x carefully, we may find the coefficients easily.

$$x^2 + 3x + 1 = A_1(x - 1)(x + 1) + A_2x(x + 1) + A_3x(x - 1)$$

$$x = 0 \rightarrow 1 = -A_1 \rightarrow A_1 = -1$$

$$x = 1 \rightarrow 5 = 2A_2 \rightarrow A_2 = \frac{5}{2}$$

$$x = -1 \rightarrow -1 = 2A_3 \rightarrow A_3 = \frac{-1}{2}$$

Now, take the constants, substitute them into the partial fractions, and replace the rational function and evaluate the integral,

$$\begin{aligned} I &= \int \frac{x^2 + 3x + 1}{x^3 - x} dx \\ &= \int \left(\frac{-1}{x} + \frac{\frac{5}{2}}{x-1} + \frac{\frac{-1}{2}}{x+1} \right) dx \\ &= \int \frac{-1}{x} dx + \int \frac{\frac{5}{2}}{x-1} dx + \int \frac{\frac{-1}{2}}{x+1} dx \\ &= -\ln x + \frac{5}{2} \ln(x-1) - \frac{1}{2} \ln(x+1) + c \end{aligned}$$

$$2. I = \int \frac{x^2+1}{x^4+2x^3+x^2} dx$$

Factor out $x^4 + 2x^3 + x^2$,

$$x^4 + 2x^3 + x^2 = x^2(x^2 + 2x + 1) = x^2(x + 1)^2$$

Put it in the denominator,

$$\begin{aligned} I &= \int \frac{x^2 + 1}{x^4 + 2x^3 + x^2} dx \\ &= \int \frac{x^2 + 1}{x^2(x + 1)^2} dx \end{aligned}$$

Apply the partial fraction decomposition,

$$\begin{aligned} \frac{x^2 + 1}{x^2(x + 1)^2} &= \\ &= \frac{A_1}{x} + \frac{A_2}{x^2} + \frac{A_3}{x + 1} + \frac{A_4}{(x + 1)^2} \\ &= \frac{A_1x(x + 1)^2 + A_2(x + 1)^2 + A_3x^2(x + 1) + A_4x^2}{x^2(x + 1)^2} \end{aligned}$$

Equate the numerators from both sides to have the main equation that needs to be solved to determine the coefficients.

$$x^2 + 1 = A_1x(x + 1)^2 + A_2(x + 1)^2 + A_3x^2(x + 1) + A_4x^2$$

Method One (equating coefficients),

$$\begin{aligned} x^2 + 1 &= A_1x(x+1)^2 + A_2(x+1)^2 + A_3x^2(x+1) + A_4x^2 \\ &= A_1x(x^2 + 2x + 1) + A_2(x^2 + 2x + 1) + A_3(x^3 + x^2) + A_4x^2 \\ &= (A_1x^3 + 2A_1x^2 + A_1x) + (A_2x^2 + 2A_2x + A_2) + (A_3x^3 + A_3x^2) + A_4x^2 \\ &= (A_1 + A_3)x^3 + (2A_1 + A_2 + A_3 + A_4)x^2 + (A_1 + 2A_2)x + A_2 \end{aligned} \quad (1)$$

$$2A_1 + A_2 + A_3 + A_4 = 1 \quad (2)$$

$$A_1 + 2A_2 = 0 \quad (3)$$

$$A_2 = 1 \quad (4)$$

Now, substitute (4) into (3), we obtain,

$$A_1 + 2 = 0 \rightarrow A_1 = -2 \quad (5)$$

and substitute (5) into (1), we obtain,

$$-2 + A_3 = 0 \rightarrow A_3 = 2 \quad (6)$$

substitute (4), (5), (6), into (2), we obtain

$$2(-2) + 1 + 2 + A_4 = 1 \rightarrow -4 + 3 + A_4 = 1 \rightarrow A_4 = 1 + 4 - 3 = 2$$

Method Two (substitution),

$$x^2 + 1 = A_1x(x+1)^2 + A_2(x+1)^2 + A_3x^2(x+1) + A_4x^2$$

$$x = 0 \rightarrow 1 = A_2$$

$$x = -1 \rightarrow 2 = A_4$$

$$x = 1 \rightarrow 2 = 4A_1 + 4 + 2A_3 + 2$$

$$\rightarrow 2 - 6 = 4A_1 + 2A_3$$

$$\rightarrow -4 = 4A_1 + 2A_3 \quad (1)$$

$$x = 2 \rightarrow 5 = 18A_1 + 9 + 12A_3 + 8$$

$$\rightarrow 5 - 17 = 18A_1 + 12A_3$$

$$\rightarrow -12 = 18A_1 + 12A_3 \quad (2)$$

from (1) and (2), we have,

$$6A_1 = -12 \rightarrow A_1 = -2 \quad (3)$$

and from (3) into (1),

$$-4 = 4(-2) + 2A_3 \rightarrow 8 - 4 = 2A_3$$

$$\rightarrow 4 = 2A_3$$

$$\rightarrow A_3 = 2$$

Now, take the constants, substitute them into the partial fractions, and replace the rational function and evaluate the integral,

$$\begin{aligned} I &= \int \frac{x^2 + 1}{x^4 + 2x^3 + x^2} dx \\ &= \int \left(\frac{-2}{x} + \frac{1}{x^2} + \frac{2}{x+1} + \frac{2}{(x+1)^2} \right) dx \\ &= \int \frac{-2}{x} dx + \int \frac{1}{x^2} dx + \int \frac{2}{x+1} dx + \int \frac{2}{(x+1)^2} dx \\ &= -2 \ln x - \frac{1}{x} + 2 \ln(x+1) - 2 \frac{1}{x+1} + c \end{aligned}$$

$$3. I = \int \frac{x^2}{x^3 - x^2 - x + 1} dx \text{ H.W}$$

$$4. I = \int \frac{-2x+4}{(x-1)^2(x^2+1)} dx \text{ H.W}$$

$$5. I = \int \frac{x^5 - x^4 - 3x + 5}{x^4 - 2x^3 + 2x^2 - 2x + 1} dx \text{ H.W}$$

$$6. I = \int \frac{x^2 + 4}{x^4 + 2x^3 - 5x^2 - 6x} dx \text{ H.W}$$